

Multiple Choice (10 marks)

1. What is the density of fat free mass?
 - (a) 0.900 g/cm³
 - (b) 0.730 g/cm³
 - (c) 1.100 g/cm³
 - (d) None of the above
2. Molecules of natural polyunsaturated fatty acids in vegetable oils contain:
 - (a) 18 carbon atoms with one carbon-carbon double bond in the cis configuration
 - (b) 20 carbon atoms with at least two carbon-carbon double bonds in the cis configuration
 - (c) 18 carbon atoms with at least two carbon-carbon double bonds in the trans configuration
 - (d) 18 carbon atoms with at least two carbon-carbon double bonds in the cis configuration
3. The thermic effect of food
 - (a) is substantially higher for carbohydrate than for protein
 - (b) is accompanied by a slight decrease in body core temperature.
 - (c) is partly related to sympathetic activity stimulation in the postprandial phase
 - (d) is not attenuated by food malabsorption.
4. Which of the following does not occur with increasing age?
 - (a) A reduction in lean body mass
 - (b) A reduction in bone density
 - (c) An increased appetite
 - (d) Impaired immune function
5. What are the main enzyme systems for catalysing ethanol to acetaldehyde?
 - (a) Alcohol dehydrogenase (ADH)
 - (b) Catalase
 - (c) Alcohol dehydrogenase (ADH), cytochrome P₄₅₀2E₁ (CYP 2 E₁), catalase
 - (d) cytochrome P₄₅₀2E₁ (CYP 2 E₁)

True / False (10 marks)

Answer True or False

6. True or False: Lactose is always the cause of milk intolerance in all individuals.
7. True or False: Macrobiotic and vegan diets are generally low in sugar content.
8. True or False: Additional dietary protein can stimulate the production of insulin-like growth hormone, which supports growth and muscle development.
9. True or False: Cholesterol and fatty acids absorbed from the intestine are packaged into chylomicrons and secreted into the lymph.
10. True or False: Pregnancy and lactation can result in bone mineral loss from the skeleton due to increased calcium demands.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of pepsinogen and gastric lipase in the digestive process within the stomach, and explain how they contribute to digestion.
12. Discuss the role of enteropeptidase in the activation of digestive enzymes in the pancreas after a meal, and explain its significance in the digestive process.

End of Paper